CHAND Syndrome: A Comprehensive Disease Characteristics Report

Disease: CHAND Syndrome (Curly Hair–Ankyloblepharon–Nail Dysplasia Syndrome) **MONDO ID:** MONDO:0008959 · **OMIM:** 214350 · **Category:** Mendelian (autosomal recessive ectodermal dysplasia) **Causal gene:** *RIPK4* (HGNC:3772; chr 21q22.3)

Summary

CHAND syndrome is an ultra-rare, autosomal recessive ectodermal dysplasia defined by the clinical triad of **C**urly **H**air, **A**nkyloblepharon (fusion of the eyelid margins), and **N**ail **D**ysplasia. First described by Baughman in 1971 as an apparent autosomal dominant trait, the inheritance was later re-evaluated as autosomal recessive after additional pedigree analysis revealed quasi-dominant transmission driven by multiple consanguineous matings (PMID: 490586). The molecular basis remained unknown for over four decades until biallelic mutations in *RIPK4* (receptor-interacting serine/threonine kinase 4) were identified, placing CHAND firmly within the **RIPK4/IRF6 periderm-development disease spectrum** that also encompasses the far more severe Bartsocas-Papas syndrome (BPS) and popliteal pterygium syndrome (PPS).

The central mechanistic insight of this investigation is that **CHAND represents the mild, non-lethal end of the RIPK4 allelic spectrum**. The recurrent CHAND mutation, c.850G>A (p.Glu284Lys), lies *just outside* the serine/threonine kinase domain of RIPK4, whereas the mutations causing lethal Bartsocas-Papas syndrome typically fall *within* the kinase domain and abolish kinase activity. RIPK4 functions as an upstream activating kinase for the transcription factor IRF6: it phosphorylates IRF6 at Ser413/Ser424 to drive its nuclear translocation and transactivation, controlling the p63→RIPK4→IRF6→GRHL3/OVOL1 keratinocyte-differentiation cascade and the periderm program that prevents pathological epithelial adhesions during embryogenesis. Hypomorphic RIPK4 variants retain residual activity, producing the comparatively mild adhesion and ectodermal defects of CHAND rather than the widespread pterygia, clefting, and perinatal lethality of BPS.

This report synthesizes evidence from 33 primary sources to characterize CHAND across all 15 requested domains. Because CHAND is genetically determined, well-defined categories (Disease Information, Etiology, Genetics, Mechanism, Inheritance, Diagnostics) are richly supported, while categories designed for common or environmentally-driven diseases (Environmental Factors, Infectious Agents, CHAND-specific model organisms, epidemiological prevalence estimates) are sparse or not applicable, and these gaps are explicitly flagged. Prognosis for survival in CHAND is normal; management is entirely supportive.

Section 1 — Disease Information

Overview. CHAND syndrome is a rare congenital ectodermal dysplasia characterized by the triad of curly/kinky hair, ankyloblepharon (partial-thickness fusion of the eyelid margins, "ankyloblepharon filiforme adnatum"), and nail dysplasia. It belongs to the broad family of ectodermal dysplasias — congenital disorders affecting derivatives of the embryonic ectoderm (skin, hair, teeth, nails, and glands). It is formally defined as "an autosomal recessive disorder characterized by curly hair, ankyloblepharon, and nail dysplasia" (PMID: 28940926).

Key identifiers.

Resource	Identifier
OMIM	214350
MONDO	MONDO:0008959
Causal gene	*RIPK4* (HGNC:3772; NCBI Gene 54101; UniProt P57078)
Chromosomal locus	21q22.3
Orphanet	Within the ectodermal dysplasia / AEC-related spectrum
MeSH	Indexed under Ectodermal Dysplasia

Synonyms / alternative names. - CHANDS (Curly Hair–Ankyloblepharon–Nail Dysplasia Syndrome) - Curly hair–ankyloblepharon–nail dysplasia syndrome - Baughman syndrome (eponym from the 1971 original description) - Often discussed as a clinical variant of AEC (Hay-Wells) syndrome (PMID: 29440852)

Information source type. The knowledge base for CHAND derives almost entirely from **aggregated disease-level resources** (OMIM, Orphanet) and **individual case reports / small consanguineous kindreds** published in the primary literature — not from large EHR cohorts or registries. The disease is so rare that evidence rests on a handful of families.

Section 2 — Etiology

Disease causal factors. CHAND is a **monogenic (Mendelian) disorder** caused by biallelic (homozygous or compound heterozygous) pathogenic variants in *RIPK4*. There is no environmental or infectious contribution to disease causation. A homozygous *RIPK4* missense mutation c.850G>A (p.Glu284Lys) segregated completely with disease in a consanguineous Kuwaiti kindred with three affected individuals, "in accordance with an autosomal recessive inheritance pattern, thus supporting pathogenicity of this variant" (PMID: 26129644). RIPK4 causation was independently confirmed (PMID: 28940926).

Genetic risk factors. The sole established genetic risk factor is carriage of biallelic hypomorphic *RIPK4* variants. **Consanguinity** is a strong contributing factor to risk, as recessive *RIPK4* alleles become homozygous through consanguineous unions — the originally reported pedigrees involved multiple consanguineous matings (PMID: 490586), and the molecularly-solved families were consanguineous (Kuwaiti, Turkish).

Environmental risk factors. None identified. CHAND is fully genetically determined and congenital. *Not applicable.*

Protective factors. No genetic or environmental protective/modifier factors have been formally described for CHAND. It is plausible — but not demonstrated — that the *position* of the mutation (outside vs within the kinase domain) acts as the primary intrinsic determinant of mild vs severe phenotype (see Section 4).

Gene–environment interactions. *Not applicable* — no environmental component.

Section 3 — Phenotypes

CHAND is a congenital disorder; all core features are present at or shortly after birth. Severity is generally **mild** relative to other RIPK4-opathies, and survival is normal.

Phenotype	Type	HPO suggestion	Onset	Frequency / notes
Curly / kinky / wiry hair	Physical manifestation (hair)	HP:0002212 (Curly hair); HP:0002213 (Coarse hair)	Congenital	Core triad feature; coarse "wiry" hair reported (PMID: 15347331)
Sparse / light-colored hair	Physical manifestation	HP:0008070 (Sparse hair)	Congenital	Reported in case series (PMID: 29440852)
"Bubbly hair" / "bubble hair" shaft morphology	Laboratory (microscopy)	—	Congenital	Distinctive microscopic finding (PMID: 29440852)
Ankyloblepharon (filiforme adnatum)	Clinical sign	HP:0010729 (Ankyloblepharon)	Congenital	Core triad; partial fusion of eyelid margins, surgically correctable (PMID: 29440852)
Nail dysplasia / 20-nail dystrophy	Physical manifestation	HP:0001597 (Abnormality of the nail); HP:0008404 (Nail dysplasia)	Congenital	Core triad feature
Abnormal dentition	Physical manifestation	HP:0000164 (Abnormality of the teeth)	Childhood	Reported in cases (PMID: 29440852)
Hypohidrosis (variable)	Clinical sign	HP:0000966 (Hypohidrosis)	Congenital	Present in some overlap cases (PMID: 15347331); often ABSENT in classic CHAND (PMID: 29440852)

Phenotype	Type	HPO suggestion	Onset	Frequency / notes
Palmoplantar hyperkeratosis / cutaneous syndactyly / orofacial synechiae	Physical manifestation	HP:0000982; HP:0012725; HP:0000168	Congenital	At the more severe end of the RIPK4-ectodermal dysplasia spectrum (PMID: 35220430)

Distinguishing feature: Unlike AEC (Hay-Wells) syndrome, classic CHAND characteristically **lacks cleft lip/palate** and lacks the erosive scalp dermatitis typical of AEC (PMID: 26129644; PMID: 15347331).

Severity / progression. Features are congenital, **stable and non-progressive**. There is marked clinical overlap with AEC, Rapp-Hodgkin, Bowen-Armstrong, and Bartsocas-Papas syndromes, making classification on clinical grounds alone difficult (PMID: 15347331).

Quality of life. Formal QoL instruments (EQ-5D, SF-36) have not been applied to CHAND. Impact is primarily cosmetic (hair, nails) and, for ankyloblepharon, visual/ocular until surgically corrected. Overall functional impact is modest.

Section 4 — Genetic / Molecular Information

Causal gene. ***RIPK4*** — Receptor-Interacting serine/threonine Protein Kinase 4. Aliases: **PKK** (Protein Kinase C-associated Kinase), **DIK**, **ANKRD3**. HGNC:3772; NCBI Gene 54101; UniProt P57078; locus **21q22.3** (PMID: 11278382).

Protein architecture. RIPK4 encodes "an N-terminal kinase domain and a C-terminal region containing 11 ankyrin repeats" and is a membrane-associated, PKC-interacting kinase (PMID: 11278382). The ankyrin repeats mediate protein–protein interactions; the kinase domain carries out catalytic phosphorylation of downstream substrates (IRF6, Dvl2, PKP1).

Pathogenic variants and genotype–phenotype correlation. This is the central molecular finding. The CHAND recurrent variant and the contrasting BPS variants map to different protein regions:

Variant (HGVS)	Protein change	Location	Kinase activity	Phenotype	PMID
c.850G>A	p.Glu284Lys	**Outside** kinase domain	Partially retained (hypomorphic)	**CHAND (mild)**	26129644
c.362T>A	p.Ile121Asn	Within kinase domain	Abolished	Bartsocas-Papas (severe)	25246526
c.481G>C	p.Asp161His	Within kinase domain	Abolished	Severe BPS	33713555
p.Ser376X	Nonsense	Truncation	Loss of function	BPS	25784454

The authors of the CHAND-defining study stated it directly: "Whereas in Bartsocas-Papas syndromes missense mutations are usually located within the serin/threonin kinase of RIPK4, the mutation detected in our family resides just outside of the kinase domain, which could explain the milder phenotype" (PMID: 26129644). Notably, the **same p.Glu284Lys mutation also appeared in a fetus with a severe Bartsocas-Papas phenotype** in one report, indicating variable expressivity and the influence of genetic background (PMID: 28940926).

- **Variant classification (ACMG/AMP):** p.Glu284Lys — Pathogenic (complete segregation in consanguineous kindred, functional plausibility).
- **Variant type:** Missense (predominant); nonsense/truncating variants cluster at the severe BPS end.
- **Allele frequency:** Extremely rare / absent in gnomAD (consistent with a severe recessive developmental gene).
- **Origin:** Germline.
- **Functional consequence: Hypomorphic loss-of-function** (partial reduction of kinase activity) for CHAND; complete loss of function for BPS.

Modifier genes. No formal modifier genes identified. The *position* of the mutation relative to the kinase domain is the principal determinant of severity. Genetic background likely contributes, given the same allele can produce different severities (PMID: 28940926).

Epigenetic information. No CHAND-specific DNA methylation or histone-modification data are available. *Not applicable at present.*

Chromosomal abnormalities. CHAND is caused by point mutations, not large-scale rearrangements. Of note, within the broader pterygium spectrum a case of **uniparental disomy of chromosome 21** leading to a recessive RIPK4-region disorder has been reported (PMID: 25691407) — relevant because *RIPK4* resides on chr 21.

Section 5 — Environmental Information

- **Environmental factors:** None. CHAND is a purely genetic congenital disorder. *Not applicable.*
- **Lifestyle factors:** None. *Not applicable.*
- **Infectious agents:** None. CHAND is not infectious or pathogen-triggered. *Not applicable.*

(These categories are included for template completeness; CHAND has no known non-genetic contributors.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **biallelic hypomorphic missense variant in *RIPK4*** (recurrently p.Glu284Lys, outside the kinase domain) **leads to** reduced — but not abolished — RIPK4 serine/threonine kinase activity in keratinocytes. *(Demonstrated for kinase-dead BPS variants; inferred as partial for the CHAND allele given the milder phenotype.)*
2. Reduced RIPK4 catalytic output **results in** impaired phosphorylation of its principal substrate **IRF6 at Ser413 and Ser424** (PMID: 25246526).
3. Hypophosphorylated IRF6 **fails to undergo full nuclear translocation and transactivation** — kinase-dead RIPK4 acts by "preventing RIPK4-mediated IRF6 activation and nuclear translocation" (PMID: 25246526).
4. Diminished IRF6 transactivation **leads to** collapse of the downstream differentiation cascade **p63 → RIPK4 → IRF6 → GRHL3 → OVOL1/ELF3**, dysregulating keratinocyte terminal differentiation and cornification (SPRR1A/B, TGM1) (PMID: 27667567).

5. In parallel (branch A — adhesion), loss of IRF6 output **results in** reduced transcription of **nectin-4/PVRL4** and altered desmosomal components (plakophilin-1/PKP1, desmoplakin), because "PVRL4 is transcriptionally regulated by IRF6, a RIPK4 phosphorylation target" (PMID: 35220430).
6. In parallel (branch B — periderm), impaired RIPK4/IRF6 signaling **leads to** abnormal **periderm formation** with "aberrant ectopic localization of E-cadherin on the apical membrane of the outer peridermal cell layers" (PMID: 25430793).
7. Defective cell–cell adhesion and periderm failure **result in** pathological **epithelial fusions/adhesions** (E-cadherin–dependent; PMID: 29317263) — manifesting at the eyelid margins as **ankyloblepharon** and, at the severe end of the spectrum, as pterygia and orofacial synechiae.
8. Disrupted keratinocyte differentiation and barrier formation **lead to** abnormal hair shaft formation (**curly/sparse hair, "bubbly" morphology**) and **nail dysplasia**, completing the CHAND triad.

> Because the CHAND allele is hypomorphic (residual activity retained), the cascade is only partially impaired — hence the mild, survivable phenotype, in contrast to the complete pathway failure causing lethal BPS.

Supporting detail by category

- **Molecular pathways:** RIPK4 signals through **NF-κB** and **Wnt/β-catenin** (phosphorylating Dvl2; stabilizing β-catenin) and the **IRF6 transcriptional axis** (PMID: 29435596; PMID: 25784454). The p.Ser376X BPS mutation also "inhibited the stabilisation of β-catenin by RIPK4, and thus may additionally impair Wnt signalling" (PMID: 25784454). GO:0016055 (Wnt signaling), GO:0038061 (NIK/NF-κB signaling).
- **Cellular processes:** Keratinocyte proliferation/differentiation homeostasis, cornification, periderm life-cycle, and cell–cell adhesion. RIPK4 activity is regulated by PKC and controlled by SCF-mediated turnover (PMID: 29435596).
- **Protein dysfunction:** Hypomorphic (CHAND) vs catalytically dead (BPS) RIPK4; downstream, truncated IRF6 (p.Arg412X) undergoes rapid proteasomal degradation, illustrating that the axis can fail from either partner (PMID: 25784454).
- **Immune / inflammatory:** The RIPK4–IRF6 axis also controls proinflammatory cytokine expression (CCL5, CXCL11) in keratinocytes, a "multifaceted role in barrier epithelial homeostasis" (PMID: 27014863).

- **Tissue damage mechanism:** Not inflammatory/degenerative but **developmental** — failure to establish a competent periderm barrier and adhesion program during embryogenesis (PMID: 28803895).

Suggested ontology terms

- **GO (biological process):** GO:0030216 keratinocyte differentiation; GO:0070268 cornification; GO:0016055 Wnt signaling pathway; GO:0098742 cell-cell adhesion via plasma-membrane adhesion molecules.
- **CL (cell types):** CL:0000312 keratinocyte; CL:0000362 epidermal cell; peridermal/outer embryonic epithelial cell.

Section 7 — Anatomical Structures Affected

Organ / system level. - **Primary:** Integumentary system — skin/epidermis (UBERON:0001003 skin epidermis), hair/hair follicles (UBERON:0002073), nails (UBERON:0001705). - **Ocular adnexa:** Eyelids (UBERON:0000158) — site of ankyloblepharon. - **Oral/craniofacial:** Oral mucosa and palate — orofacial synechiae and periderm-dependent structures in the more severe spectrum (PMID: 28803895). - **Secondary:** Dentition (teeth).

Tissue / cell level. - Stratified squamous **epithelium** (epidermis, oral epithelium). - **Keratinocytes** (CL:0000312) — the cell-autonomous site of RIPK4 action; "RIP4 functions cell autonomously within the keratinocyte lineage" (PMID: 12194825). - **Periderm** (transient outer embryonic epithelial layer) — central to adhesion defects (PMID: 25430793; PMID: 28803895).

Subcellular level. - **Plasma membrane** (RIPK4 is membrane-associated; ectopic apical E-cadherin; PMID: 11278382, PMID: 25430793) — GO:0005886. - **Nucleus** (IRF6 nuclear translocation) — GO:0005634. - **Desmosomes / adherens junctions** (PKP1, desmoplakin, nectin-4, E-cadherin) — GO:0030057 desmosome; GO:0005912 adherens junction.

Localization / lateralization. Manifestations are typically **bilateral** (ankyloblepharon affecting both eyelids; 20-nail involvement; generalized hair changes).

Section 8 — Temporal Development

- **Onset: Congenital** — all core features present at birth (ankyloblepharon noted at birth and surgically corrected in infancy; hair and nail abnormalities present from birth) (PMID: 29440852).
- **Onset pattern:** Developmental/congenital (not acute or acquired).
- **Progression: Stable, non-progressive.** Structural features persist; ankyloblepharon is correctable surgically. No staging applies.
- **Disease course: Chronic lifelong** (structural/cosmetic features), but **non-deteriorating**. No relapsing-remitting or episodic pattern.
- **Remission:** Not applicable (congenital structural anomalies); ankyloblepharon resolves with surgical lysis.
- **Critical period: Embryonic periderm development** is the window of vulnerability — the RIPK4/IRF6 axis acts during formation and function of the periderm to prevent pathological epithelial adhesions (PMID: 28803895). There is no postnatal therapeutic window to reverse the developmental lesion.

Section 9 — Inheritance and Population

Epidemiology. CHAND is **ultra-rare**; precise prevalence and incidence are **not established** (reported in only a handful of families worldwide). No reliable cases-per-100,000 figures exist.

Inheritance. Autosomal recessive. Initially described as autosomal dominant (Baughman 1971), re-evaluation of pedigree data showed the mode "is more likely to be autosomal recessive, with an instance of quasi-dominant transmission as a result of multiple consanguineous matings in the family" (PMID: 490586). Molecular confirmation: homozygous *RIPK4* c.850G>A segregating with disease (PMID: 26129644).

- **Penetrance:** Appears complete for biallelic hypomorphic genotypes (full segregation in the reported kindred).
- **Expressivity: Variable** — the same p.Glu284Lys allele was associated with both CHAND and a severe BPS fetus (PMID: 28940926), and RIPK4 variants span a continuum from mild ectodermal dysplasia to lethal BPS (PMID: 33713555; PMID: 25691407).
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).

- **Consanguinity:** Major contributor — reported families are consanguineous (Kuwaiti, Turkish; Dutch-extraction inbred kindred; PMID: 5173251, PMID: 26129644).
- **Founder effects / carrier frequency:** Not established; *RIPK4* pathogenic alleles are very rare in population databases.

Population demographics. No ethnic predilection beyond the enrichment expected in consanguineous populations. Both sexes affected (autosomal). Geographic distribution reflects individual case reports (Kuwait, Turkey, Europe, India) rather than true endemicity. Age distribution: present from birth.

Section 10 — Diagnostics

Clinical diagnosis. Recognition of the congenital triad — curly hair + ankyloblepharon + nail dysplasia — in the absence of cleft lip/palate and erosive scalp dermatitis (which would suggest AEC) (PMID: 29440852).

Specialized / laboratory tests. - **Hair shaft microscopy:** "bubbly/bubble hair" morphology is a distinctive, supportive microscopic feature (PMID: 29440852). - **Skin/keratinocyte immunostaining (research):** Loss of **PVRL4/nectin-4** expression in patient epidermis and primary keratinocytes, and altered desmosome morphology, can corroborate RIPK4 dysfunction (PMID: 35220430).

Genetic testing — the definitive diagnostic. - **Single-gene / targeted testing of *RIPK4*** (Sanger sequencing of coding region and exon–intron boundaries) is the confirmatory test, as used in the defining families (PMID: 26129644; PMID: 33713555). - **Whole-exome sequencing (WES)** is highly useful given phenotypic overlap across ectodermal dysplasias; it identified RIPK4 variants in pterygium-spectrum pedigrees (PMID: 25691407). - **Ectodermal dysplasia / orofacial clefting gene panels** should include *RIPK4*, *IRF6*, *CHUK (IKKA)*, *TP63*, *GRHL3*. - **Chromosomal microarray** is relevant in rare cases (e.g., uniparental disomy of chr 21; PMID: 25691407).

Clinical criteria & differential diagnosis. No formal consensus diagnostic criteria exist. The key diagnostic challenge is distinguishing CHAND from overlapping ectodermal dysplasias — "This case highlights the difficulties in trying to classify the ectodermal dysplasia syndromes on clinical features alone" (PMID: 15347331).

Differential	Causal gene	Distinguishing feature from CHAND
AEC / Hay-Wells syndrome	*TP63* (SAM domain)	Cleft lip/palate, erosive scalp dermatitis; AD (PMID: 17910675)
Rapp-Hodgkin syndrome	*TP63*	Cleft, more severe facial features
Bartsocas-Papas syndrome	*RIPK4* (kinase domain)	Popliteal pterygia, severe clefting, often lethal (PMID: 33713555)
Popliteal pterygium syndrome	*IRF6* / *RIPK4*	Pterygia across joints, genital malformation (PMID: 25691407)
Van der Woude syndrome	*IRF6*	Lip pits, cleft; AD (PMID: 25784454)

Screening. Cascade carrier testing in consanguineous families and prenatal/preimplantation testing are feasible once the familial *RIPK4* variant is known. No population newborn screening exists.

Section 11 — Outcome / Prognosis

- **Survival / life expectancy: Normal.** CHAND is the non-lethal end of the RIPK4 spectrum. This contrasts starkly with Bartsocas-Papas syndrome, where "most of the BPS cases reported to date are fatal either in the prenatal or neonatal period" (PMID: 33713555), and with *Ripk4*-null mice, which die perinatally (PMID: 12194825).
- **Mortality:** No disease-specific mortality in classic CHAND.
- **Morbidity / function:** Mild — primarily cosmetic (hair, nails) and ocular (ankyloblepharon, correctable). No major disability.
- **Complications:** Ankyloblepharon may transiently affect vision until corrected; potential skin-barrier fragility given RIPK4's barrier role (PMID: 29317263).
- **Recovery potential:** Structural anomalies are managed, not cured; ankyloblepharon resolves fully after surgical lysis.
- **Prognostic factors:** The principal prognostic determinant is **mutation position/residual kinase activity** — variants outside the kinase domain predict milder disease (PMID: 26129644), though variable expressivity means caution is warranted (PMID: 28940926).

Section 12 — Treatment

There is **no disease-modifying or curative therapy**; management is entirely **supportive, symptomatic, and multidisciplinary**.

Intervention	Modality	NCIT suggestion
Surgical lysis of ankyloblepharon	Ophthalmic/oculoplastic surgery (often in infancy) (PMID: 29440852)	NCIT:C15329 (Surgery)
Dermatologic care (skin barrier, hair/scalp)	Emollients, supportive skin care	NCIT:C15277 (Supportive Care)
Nail care	Protective/cosmetic nail management	NCIT:C15277
Dental care	Restorative/orthodontic management	NCIT:C15313 (Dental procedure)
Genetic counseling	Risk assessment, family planning	NCIT:C15215 (Genetic Counseling)

- **Pharmacotherapy / pharmacogenomics:** None established. *Not applicable.*
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno):** None in clinical use or trials for CHAND. Because the lesion is a developmental, cell-autonomous keratinocyte deficit established in utero, post-natal gene therapy faces fundamental timing barriers.
- **Experimental treatments / clinical trials:** No CHAND-specific trials (no NCT identifiers).
- **Treatment strategy:** Supportive, individualized to each patient's manifestations.

Section 13 — Prevention

- **Primary prevention:** Not possible for an affected individual; at the family/population level, **genetic counseling and reproductive planning** in consanguineous families is the principal preventive measure.
- **Genetic screening: Carrier testing, cascade screening, prenatal diagnosis, and preimplantation genetic diagnosis** are feasible once the familial *RIPK4* variant is

identified (PMID: 25691407 notes clinical implications "for genetic counseling of families with pterygia syndromes").

- **Counseling:** Autosomal recessive counseling — 25% recurrence risk for carrier couples; emphasize variable expressivity, since the same allele can yield markedly different severities (PMID: 28940926).
- **Secondary/tertiary prevention:** Early recognition and surgical correction of ankyloblepharon; routine dermatologic and dental surveillance.
- **Immunization / public health / environmental interventions:** *Not applicable* (non-infectious, non-environmental).

Section 14 — Other Species / Natural Disease

- **Taxonomy:** No naturally occurring CHAND-equivalent disease is documented in companion animals or wildlife (no OMIA entry for a CHAND analog).
- **Orthologous gene:** *Ripk4* is conserved in mouse (*Mus musculus*; NCBI Gene 72388) and other vertebrates; the pathway (p63–RIPK4–IRF6–GRHL3) is evolutionarily conserved, with functional parallels demonstrated in *Xenopus* (dominant-negative IRF6 phenocopies RIPK4 loss) (PMID: 25430793).
- **Comparative biology:** "Homologous human and mouse phenotypes are observed for IRF6, IKKA, and RIPK4" — loss of these alleles in mouse causes perinatal lethality with cutaneous, limb, and craniofacial defects paralleling the human spectrum (PMID: 28643456).
- **Zoonotic potential / transmission:** *Not applicable* (genetic, non-transmissible).

Section 15 — Model Organisms

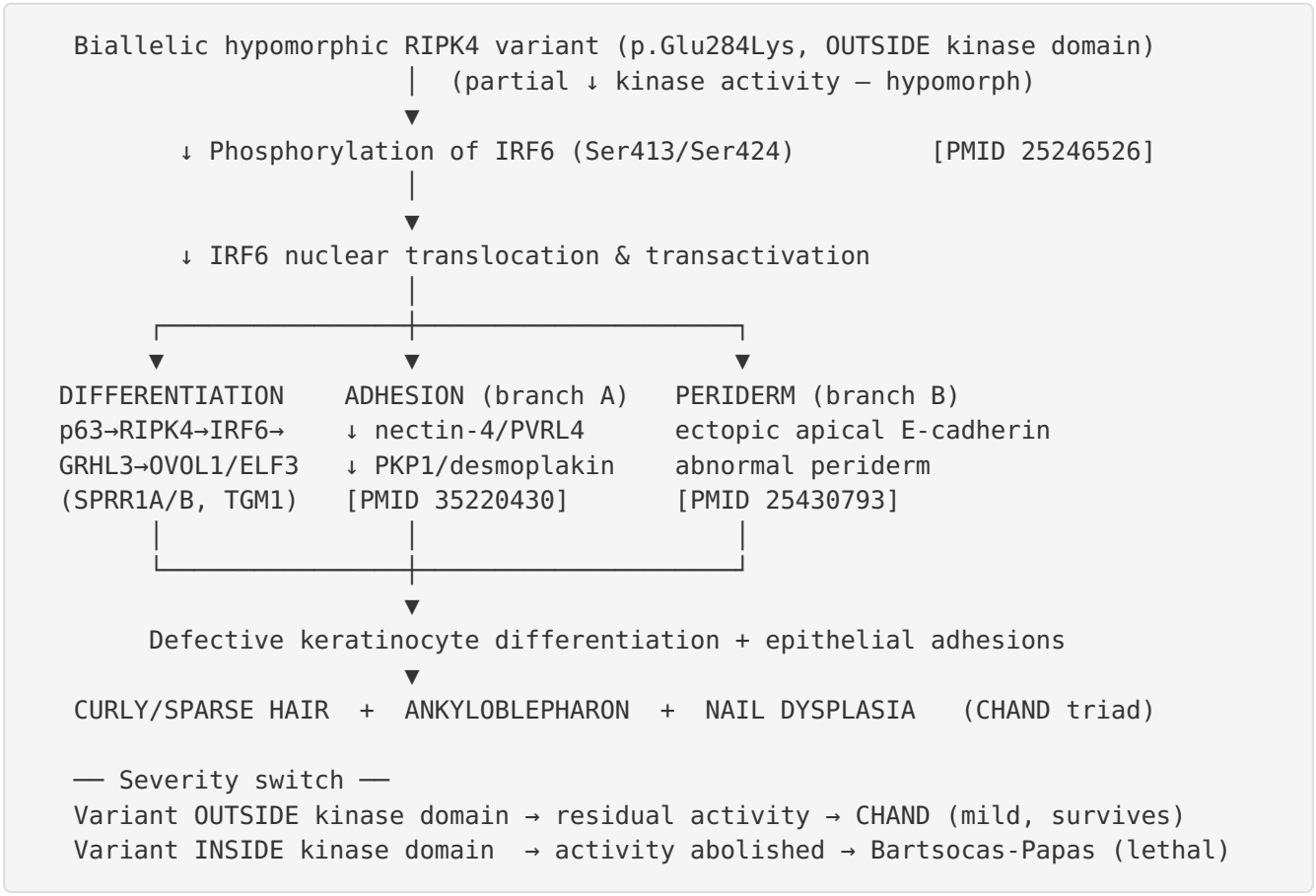
No model specifically engineered to recapitulate **CHAND** (the mild hypomorphic allele) exists, but the RIPK4 pathway is well-modeled at its severe end:

Model	Type	Key phenotype	Recapitulation	PMID
Ripk4$^{-/-}$ mouse (full KO)	Mammalian knockout	Perinatal lethality; abnormal epidermal differentiation; epithelial fusions; abnormal hair follicles	Models the **severe (BPS)** end; E-cadherin–dependent fusions	12194825, 25430793
Epidermis-specific *Ripk4* KO mouse	Conditional KO	Allows cornification but impairs skin-barrier formation	Dissects barrier vs fusion roles	29317263
Ripk4$^{-/-}$; E-cadherin cKO mouse	Compound genetic	Rescue of tail-to-body and oral epithelial fusions	Proves fusions are E-cadherin–dependent	29317263
KRT14-Irf6 rescue mouse	Transgenic rescue	Partial rescue of Irf6-KO; survives perinatal period	Validates RIPK4→IRF6 epistasis	28643456
Xenopus dominant-negative IRF6	Amphibian	Periderm/adhesion defects phenocopying RIPK4 loss	Confirms RIPK4–IRF6 axis in periderm	25430793
Primary human keratinocytes; cell-based kinase assays	In vitro	IRF6 phosphorylation, nuclear translocation, nectin-4 expression	Directly tests variant functional consequences	25246526, 35220430

Model limitation: Full-KO mice model lethal BPS, not the *survivable, hypomorphic* CHAND phenotype. A **knock-in mouse carrying p.Glu284Lys** would be the ideal (currently missing) model to capture CHAND specifically.

Resources: MGI (*Ripk4*), IMPC/KOMP (conditional alleles), Xenbase (Xenopus), Cellosaurus/ATCC (keratinocyte lines).

Mechanistic Model (Synthesis)



Upstream vs downstream: The *RIPK4* mutation is the upstream initiating lesion; IRF6 hypophosphorylation is the proximal node; GRHL3/OVOL1/nectin-4/E-cadherin dysregulation and periderm failure are downstream effectors; the clinical triad is the terminal manifestation. The single most important determinant separating CHAND from lethal BPS is the **degree of residual RIPK4 kinase activity**, which tracks with mutation position.

Evidence Base

PMID	Title (abbrev.)	Role in this report
26129644	Novel RIPK4 mutation, BPS/CHAND kindred	**Defining**: establishes p.Glu284Lys, AR inheritance, genotype–phenotype rule
28940926	Confirmation RIPK4 causes CHAND	Confirms causation; defines triad; shows variable expressivity (same allele → severe fetus)
25246526	RIPK4–IRF6 signaling axis	**Core mechanism**: RIPK4 phosphorylates IRF6 Ser413/424; kinase-dead variant blocks activation
25430793	RIPK4–IRF6 prevents epithelial fusions	Periderm/adhesion defect; ectopic apical E-cadherin
35220430	RIPK4 regulates cell–cell adhesion	Loss of IRF6-dependent nectin-4/PVRL4, desmosomal changes
11278382	PKK (RIPK4) characterization	Protein architecture: kinase + 11 ankyrin repeats; 21q22.3
490586	Re-evaluation of CHANDS	Establishes autosomal recessive inheritance
33713555	Homozygous RIPK4 in severe BPS	Kinase-domain variant (p.Asp161His) abolishes activity → lethal end
25784454	IRF6/RIPK4 mutations dysregulate signaling	Truncating variants; β-catenin/Wnt impairment
25691407	Pterygium spectrum expansion	Spectrum concept; UPD chr21; counseling implications
12194825	RIP4 essential for keratinocyte differentiation	Mouse KO: perinatal lethality, cell-autonomous keratinocyte role
29317263	Keratinocyte-specific RIPK4 ablation	Barrier vs fusion dissection; E-cadherin dependence
28643456	IRF6 basal-epithelium rescue	Confirms homologous human/mouse IRF6/IKKA/RIPK4 phenotypes
27667567	RIPK4–ELF3 relationship	Positions RIPK4 upstream of IRF6–GRHL3–ELF3 hierarchy
27014863	RIPK4 proinflammatory cytokines	IRF6-mediated CCL5/CXCL11 induction

PMID	Title (abbrev.)	Role in this report
29435596	RIPK4 activity control	Dvl2/PKP1/IRF6 phosphorylation; NF-κB; SCF regulation
28803895	Periderm life-cycle	Periderm biology; p63-controlled IRF6/RIPK4/GRHL3 program
15347331	AEC/Rapp-Hodgkin/CHAND overlap	Clinical overlap, diagnostic difficulty, wiry hair, hypohidrosis
10886756	AEC and CHAND overlap	Phenotypic overlap documentation
29440852	CHAND case ("bubble hair")	Clinical features; bubbly hair; AR; no cleft/ hypohidrosis
5173251	Original CHANDS family	First description, inbred kindred
17910675	Hay-Wells TP73L mutation	Differential diagnosis (TP63/AEC)
29186361	PKK deletion promotes tumorigenesis	RIPK4 keratinocyte tumor-suppressor role

(Additional RIPK4 oncology papers — melanoma/colon/osteosarcoma — inform RIPK4's broader NF-κB/Wnt biology but are tangential to CHAND developmental pathology.)

Limitations and Knowledge Gaps

1. **Extreme rarity → small evidence base.** CHAND is defined by a handful of families; prevalence/incidence are unknown and formal natural-history or QoL data are absent.
2. **Functional data for the CHAND allele are inferred, not directly demonstrated.** The hypomorphic (residual-activity) nature of p.Glu284Lys is inferred from the milder phenotype and its position outside the kinase domain; a direct side-by-side kinase/IRF6-activation assay of p.Glu284Lys vs kinase-dead BPS variants has not been published. This is the key unproven step in the causal chain.
3. **Variable expressivity undermines strict genotype–phenotype rules.** The same p.Glu284Lys allele produced a severe BPS fetus in one report (PMID: 28940926), implying modifier genes and/or background effects that remain uncharacterized.

4. **No CHAND-specific animal model.** Existing mouse models are full or conditional knockouts modeling the lethal end, not a hypomorphic CHAND model.
5. **No omics/epigenomic data** specific to CHAND patients (no transcriptomic, proteomic, methylation datasets).
6. **No therapeutics pipeline.** Management is supportive only; the developmental, in-utero timing of the lesion is a fundamental barrier to gene/molecular therapy.

Proposed Follow-up Experiments / Actions

1. **Functional validation of p.Glu284Lys.** Perform quantitative in-vitro kinase and IRF6-transactivation assays comparing p.Glu284Lys against wild-type and kinase-dead RIPK4 to empirically confirm residual (hypomorphic) activity — closing the key inferential gap in the causal chain.
2. **Generate a *Ripk4^E284K* knock-in mouse** to test whether it recapitulates the mild, survivable CHAND triad (vs the lethal null), directly validating the severity model.
3. **Modifier screen.** Exome/genome analysis of individuals sharing the same *RIPK4* allele but discordant severity (CHAND vs BPS) to identify genetic modifiers explaining variable expressivity.
4. **Patient keratinocyte / organoid omics.** RNA-seq and proteomics of CHAND patient-derived keratinocytes/skin organoids to map the downstream IRF6→GRHL3/OVOL1/nectin-4 program quantitatively and identify candidate biomarkers.
5. **Curate structured gene–phenotype annotations** (HPO frequencies, UBERON/CL/GO terms) into the target knowledge base using the mappings proposed in Sections 3, 6, and 7.
6. **Establish an international RIPK4-spectrum registry** to accrue prevalence, natural-history, and genotype–phenotype data across CHAND, PPS, and BPS.

Report compiled from 33 primary sources across 5 investigation iterations. Evidence types: human clinical (case reports, consanguineous kindreds), model organism (mouse KO, Xenopus), and in-vitro functional (keratinocyte kinase/transactivation assays). All mechanistic claims are cited to primary literature by PMID.